

Jesus Sanchez

Engineering Portfolio

Outline

1

About Me

2

Skills

3

NASA JPL

4

Liquid Propulsion Lab

5

Combustor Research

6

Coursework

7

Contact Information

About Me

My name is Jesus Sanchez, Aerospace Engineer from University of Southern California, with experience in propulsion and mechanical design.

I am motivated by the opportunity to solve complex challenges and contribute to advancements in technology that have a tangible impact on humanity's future. I am particularly inspired by purpose-driven work. Knowing that my work will directly contribute to groundbreaking technology motivates me to excel in every project I undertake.



Skills

CAD, CFD, FEA

- Siemens NX
- SOLIDWORKS
(+Simulation)
- Ansys Fluent
- Ansys Mechanical
- OpenFOAM

Software

- MATLAB
- LabView
- Python
- Simulink
- Cantera
- Teamcenter

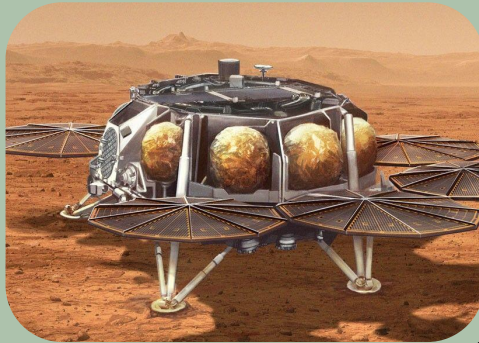
Technical

- GD&T
- Tolerance Analysis
- 3D Printing

NASA JPL

(May 2022 - Feb 2024)

During my internship, I worked as a Mechanical Engineer for the Electronic Packaging group. I began by designing ground support equipment (GSE) then transitioned to electromechanical systems for flight applications. Progressively, I led the design of an electromechanical unit for the Sample Retrieval Lander (SRL).



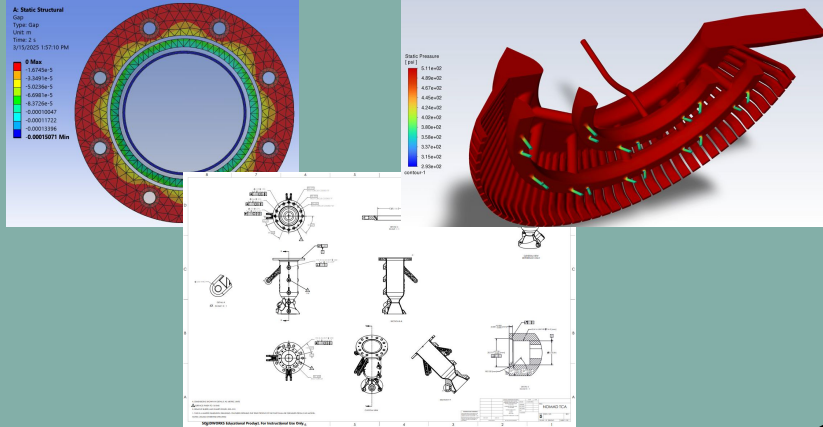
Responsibilities

- CAD (Siemens NX)
 - ◆ Design for manufacturing and assembly (DFM & DFA)
- Engineering Drawings
 - ◆ Part DWG
 - ◆ Assembly DWG
 - ◆ Interface DWG
- GD&T
- Written procedures for build and assembly
- Troubleshoot hardware build

Liquid Propulsion Lab (LPL)

(Sep 2024 - Present)

At USC's LPL, I worked on the design and testing of a regeneratively cooled 3 kN Jet-A/LOX rocket engine, with responsibilities spanning detailed design, thermal analysis, CFD, and FEA for the development of the injector, thrust chamber and nozzle.



Responsibilities

- CAD (Siemens NX)
- Injector, chamber, nozzle design
- CFD (injector, chamber and nozzle)
- MATLAB (thermal analysis, nozzle contour)
- Bolt analysis
- FEA



(May 2023 - Present)



-
- Figure 10 consists of a graph and a schematic diagram. The graph plots Air Flow Rate (L/min) on the y-axis (0 to 35) against Fuel Flow Rate (L/min) on the x-axis (0 to 10). It shows several linear relationships for different orifice diameters (ϕ) under 60% hydrogen dilution. A legend indicates that blue circles represent 'Stable' operation and open circles represent 'Unstable' operation. The schematic diagram to the right shows a burner assembly with 'Fuel' inlets at the top and an 'Air' inlet at the bottom.

Spacecraft Design

Spacecraft structures and mechanisms, space environment, orbital mechanics, attitude determination and control, power systems, thermal control

Rocket Propulsion

Liquid and solid propulsion, nozzles, propellants, feed system, flight performance, orbital mechanics & staging

Compressible Gas Dynamics

Shocks, expansions, Rayleigh flow, Fanno flow, unsteady wave motion, linearized potential theory, transonic, hypersonic flow, and CFD project

Partial Differential Equations

Solve PDE's (elliptic, parabolic, hyperbolic) in cartesian, cylindrical, spherical coordinates and time dependant

Fundamentals and Applications of Combustion

Chemical thermodynamics, chemical kinetics, transport phenomena, non-premixed, premixed flames, Cantera and FDM project

Numerical Methods

Root-finding methods, interpolation and curve fitting, numerical differentiation and integration, numerical solutions to ODE's and PDE's

Coursework

Contact me

jesusfsa@usc.edu

(805)815-7654

<https://www.linkedin.com/in/jesus-sanchez1/>

<https://jesusfsanchez.github.io/>